Numerical Method and Implementation of Highly Compliant Structures in Waves

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Introduction

The aquatic vegetation, such as seaweeds, is usually quite flexible and tends to bend following the flow. The fundamental problem involved is the interaction between waves and highly compliant structures. A large scale with numerous structures is usually of interest, for example, wave attenuation over a flexible canopy, a typical application involving massive structures, which calls for a practical and efficient solver. A linearized flow solver coupled with the truss spring model is developed for the problem.

Truss Spring Model

The truss spring model is based on the truss model, which has been widely used for net cages. An equivalent couple is applied to the two nodes supporting that truss to replace the moment due to bending. A massless rotation spring is added to each node of the truss model to model the structure bending stiffness. Later for the hydrodynamic modelling of floating seaweed farms, the longlines on which seaweeds are suspended, and other structure lines can also be modelled by the truss spring model.

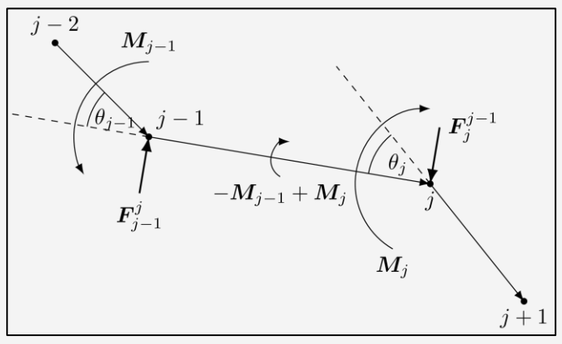

Fig. Bending momentum and the corresponding couple on a truss element

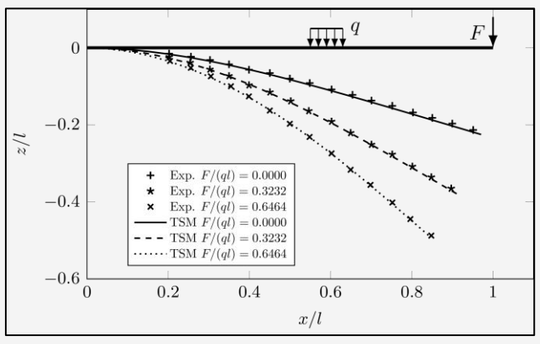

Fig. Measured and predicted deflections of a cantilever beam under self-weight and a tip load

Linearized Flow Solver

The continuity and linearized momentum equations have been widely used in the derivation of analytical solutions to wave attenuation over permeable media, which inspired us to develop a linearized flow solve. We can in this way avoid free surface tracking and keep high accuracy regarding wave propagation. The solver is quite efficient since the matrix is only decomposed initially.

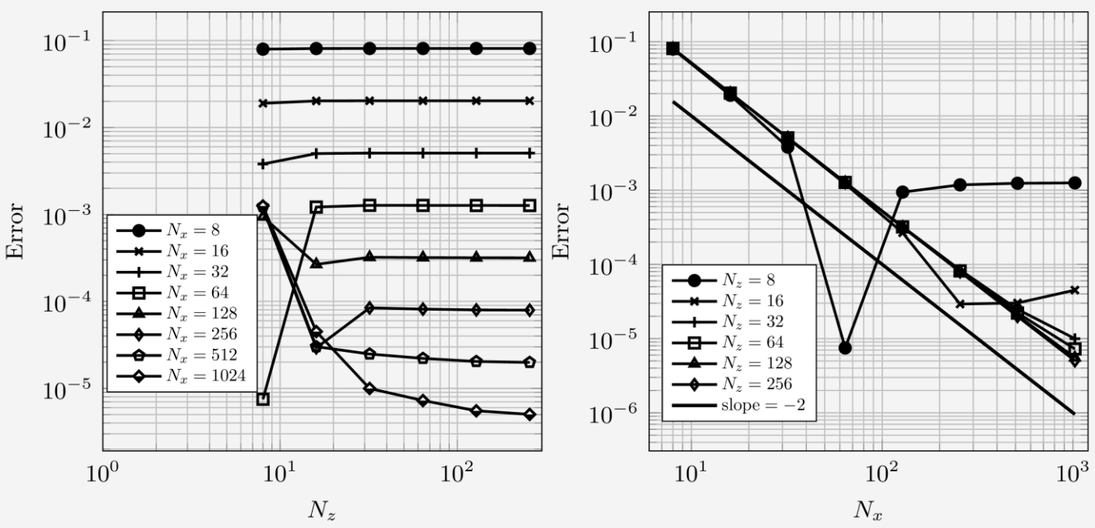

Fig. Convergence of the free surface elevation at $kh = 4$

Waves over a Rigid Canopy

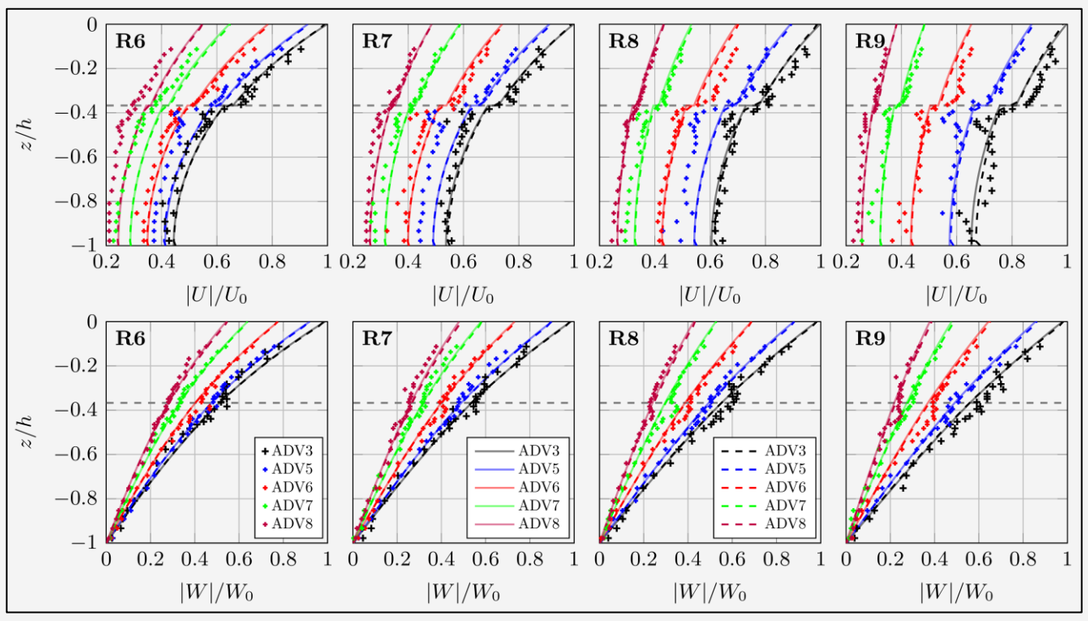

Fig. Comparison of the velocity amplitude profiles between measured data (marks), analytical solutions(solid lines), and present numerical results (dashed lines)

Single Blade in Waves

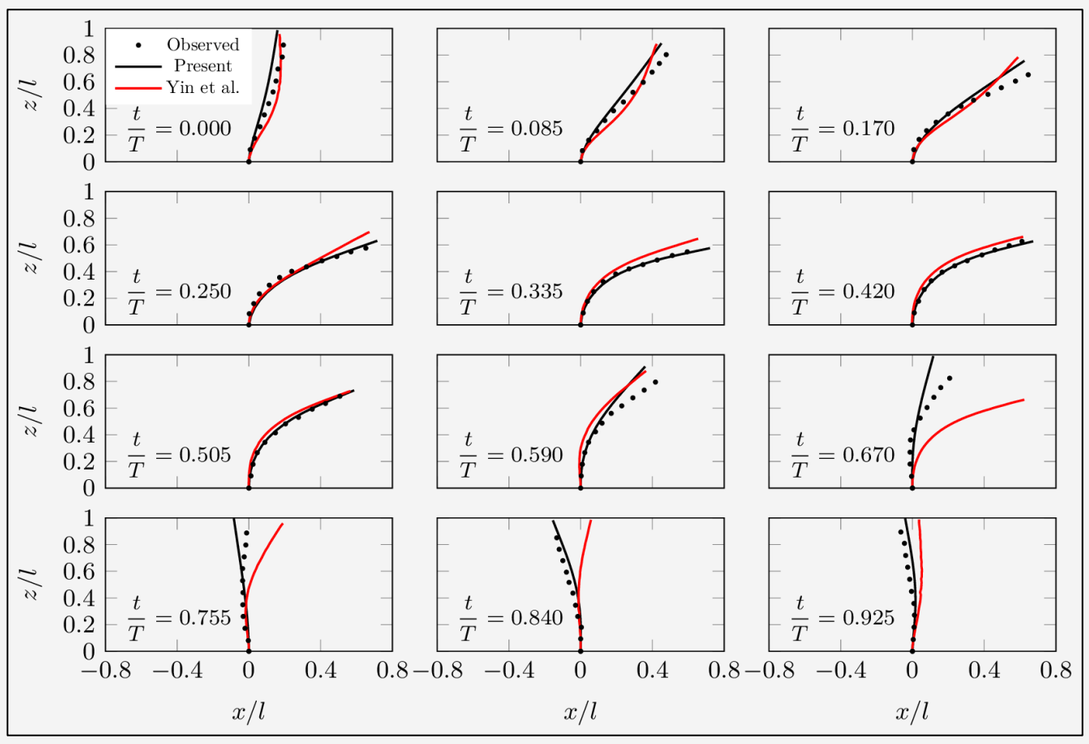

Fig. Reconfiguration of a single blade in waves over one period

Flexible Canopy in Waves

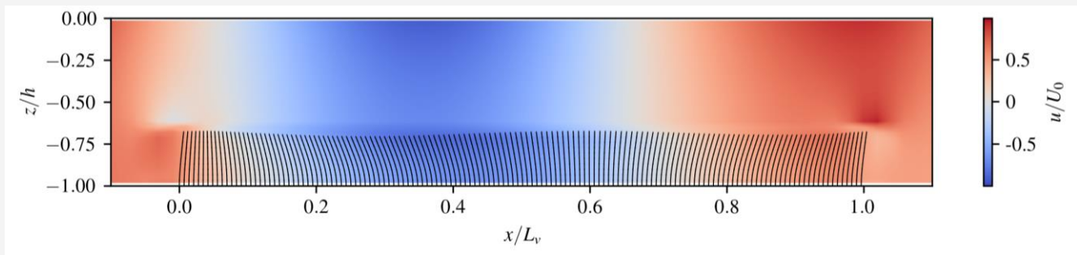

Fig. Vegetation reconfiguration (black lines) and the horizontal velocity

Ongoing

- Extend the model into 3D
- Load model: cross-flow or axial-flow principle (large KC, large Ca)
- In waves and current
- Experiments (towing tests and forced oscillations)
- Instability analysis